Severn Lortie

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Technical Skills

Languages: C/C++, Rust, Python, JavaScript
Embedded & Robotics: Microcontrollers, Device Drivers, UART, SPI, I^2C, CAN, Modbus, NMEA 2000, ROS, Motion Planning, Control
Tools: Linux, Git, CMake, Docker, Eigen

Experience

Software Engineering Intern May 2023 – August 2024
Distributive Corp. *Kingston, ON*

- Built and presented an LLM inference demo on Distributive's parallel-computing platform to Nokia's CTO and AI/HPC lead, contributing to follow-up meetings and a collaboration.
- Co-developed, from scratch, the company's Python API for its core parallel-computing platform in a two-engineer effort using an embedded JavaScript-in-Python runtime; achieved feature parity with the native JavaScript API and supported several non-trivial workloads.
- Developed a distributed-system performance dashboard that exposed web-worker bottlenecks; findings led to a fix reducing average slice scheduling time by 30 minutes.
- Designed a non-blocking WebSocket bridge in C++ from the embedded V8 workers to the Chrome DevTools Protocol; the resulting debugger was used to fix critical product bugs.
- Designed a mobile Web Worker unload-detection mechanism using JavaScript event-loop behavior; authored technical analysis for CTO and CEO review, and the accepted design greatly reduced false-positive dead-worker detection on mobile.

Undergraduate Researcher Summer 2025
Queen's University *Kingston, ON*

- Selected for a competitive $9,000 Undergraduate Student Research Award (USRA) to conduct marine robotics research integrating sensing, embedded hardware, and robotic control.
- Developed low-level NMEA 2000 / CAN drivers and a ROS bridge for commercial marine sensors, integrating navigation data into the autonomy stack for use by other researchers.

Master's Research

Autonomous Sailboat Research September 2025 – Present
Queen's University – Master's in Robotics, Electrical and Computer Engineering *Kingston, ON*

- Designed and deployed the complete control, actuation, and motion-planning stack for an autonomous sailboat; validated the system through repeated on-water field tests in Lake Ontario.

Projects

SolarStatus | *Modbus, Serial, Raspberry Pi* GitHub

- Reverse engineered Modbus messages from an EPEVER solar charge controller and implemented serial communication, checksums, and protocol handling on a Raspberry Pi.
- Built a web dashboard showing live battery state-of-charge, solar input power, charging state, and system power usage.

Education

Queen's University September 2025 – Present
Master's in Robotics, Electrical and Computer Engineering *Kingston, ON*

- Awards: Vector Scholarship in Artificial Intelligence ($17,500) and Ontario Graduate Scholarship ($15,000).

Queen's University September 2020 – May 2025
Bachelor of Computer Engineering, Electrical and Computer Engineering *Kingston, ON*

- Queen's VEX Robotics – Firmware Team: designed an RP2040 high-baud-rate interface for LiDAR and rotary encoders used for localization and mapping at the VEX U Robotics World Championship.
- Winner, Amazon HACKTO Hackathon (2022); Dean's List (2020–2023).